Arush Nandakumar Menon

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Portfolio • LinkedIn • GitHub

SUMMARY

An undergraduate in software engineering specializing in Computer Vision and Generative AI. Expertise in architecting real-time surveillance systems using YOLOv8 and deploying scalable ML pipelines with Docker and Kubernetes. Proficient in optimizing inference latency for edge applications and integrating AI into production-grade full-stack solutions. Experienced in technical leadership and Agile methodologies.

WORK EXPERIENCE

Startrit Infratech Private Limited Dec 2025 – Present
Software Intern Remote, India

- Spearheaded the architecture of scalable business solutions with a 5-member agile team, utilizing Python to increase backend throughput by 30%.
- Developed robust PostgreSQL services, securing 10k+ records via JWT authentication and optimized queries.
- Orchestrated real-time features through Socket.IO, reducing data sync latency by 40% for 500+ concurrent users.

The Electrical Vehicle Club, VIT Bhopal University Jan 2024 – May 2025
Technical Lead Sehore, India

- Directed 3 flagship technical events for 500+ students, managing logistics and a 5k+ prize pool with 100% % on-time execution.
- Revamped the official EV Club website, launching 5 new features that improved user experience and increased site traffic by 25%.

TECHNICAL SKILLS

AI / Computer Vision	YOLOv8, DeepSORT, X3D, OpenCV, TensorFlow, PyTorch, Ollama, RAG
Cloud & DevOps	Docker, Kubernetes, AWS (EC2, S3), Prometheus, Grafana, CI/CD
Languages	Python, C/C++, TypeScript, JavaScript, SQL, Java
Web Technologies	React 18, Next.js 15, Tailwind CSS, Node.js, FastAPI, REST APIs
Soft Skills	Technical Leadership, Agile Methodology, Team Management, Strategic Planning

PROJECTS

CivicGuard: AI Surveillance | *Python, YOLOv8, FastAPI, Docker* GitHub | Live | Jan 2026 – Present

- Architected Smart City surveillance integrating YOLOv8 for weapon detection with 95% recall accuracy.
- Engineered tracking pipeline using DeepSORT, processing 4 concurrent streams with sub-100ms latency.
- Deployed via Docker and Kubernetes for auto-scaling, maintaining 99.9% system uptime under load.

Student Hub - AI Suite | *Next.js, Gemini API, Prompt Engineering* GitHub | Live | Dec 2025 – Jan 2026

- Integrated Google Gemini API to automate note summarization, reducing student study planning time by 35%.
- Formulated prompt engineering workflows for parsing academic text into JSON, achieving a 98% parsing success rate.
- Maximized API efficiency by 20% via caching, achieving a 95+ Lighthouse performance score.

LocalMind: Hybrid Assistant | *Python, Ollama, Gemini, Supabase* GitHub | Live | Jul 2025 – Nov 2025

- Constructed hybrid inference architecture, routing 1,000+ daily queries between Cloud and 7+ Local Ollama models.
- Built autonomous agent system with 3+ tool integrations (Files, Calendar) for complex task execution.
- Implemented semantic memory summarization, reducing token usage by 60% and API costs by 45%.

EDUCATION

VIT Bhopal University Sep 2023 – Present
B.Tech in Computer Science Engineering | CGPA: 8.03/10.0 Sehore, India
Relevant Coursework: DSA, Operating Systems, Computer Networks, AI/ML Fundamentals

Sri Chaitanya Techno School 2022
Senior Secondary Education (Class XII) | Score: 87% Bangalore, India

ACHIEVEMENTS & CERTIFICATION

- **Certifications:** Introduction to Machine Learning (NPTEL-IIT, Apr 2025), The Bits and Bytes of Computer Networking (Google, Nov 2025), Python Basic (HackerRank, 2024), AWS Cloud Practitioner Essentials.
- **Languages:** English (Fluent), Hindi (Fluent), Malayalam (Native), Tamil (Intermediate), Kannada (Basic).